End-to-End Sustainable Supply Chain Transformation

Electronic Express GmbH

Round 2

ZENITH

ROAD TO SUSTAINABILITY

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About Us

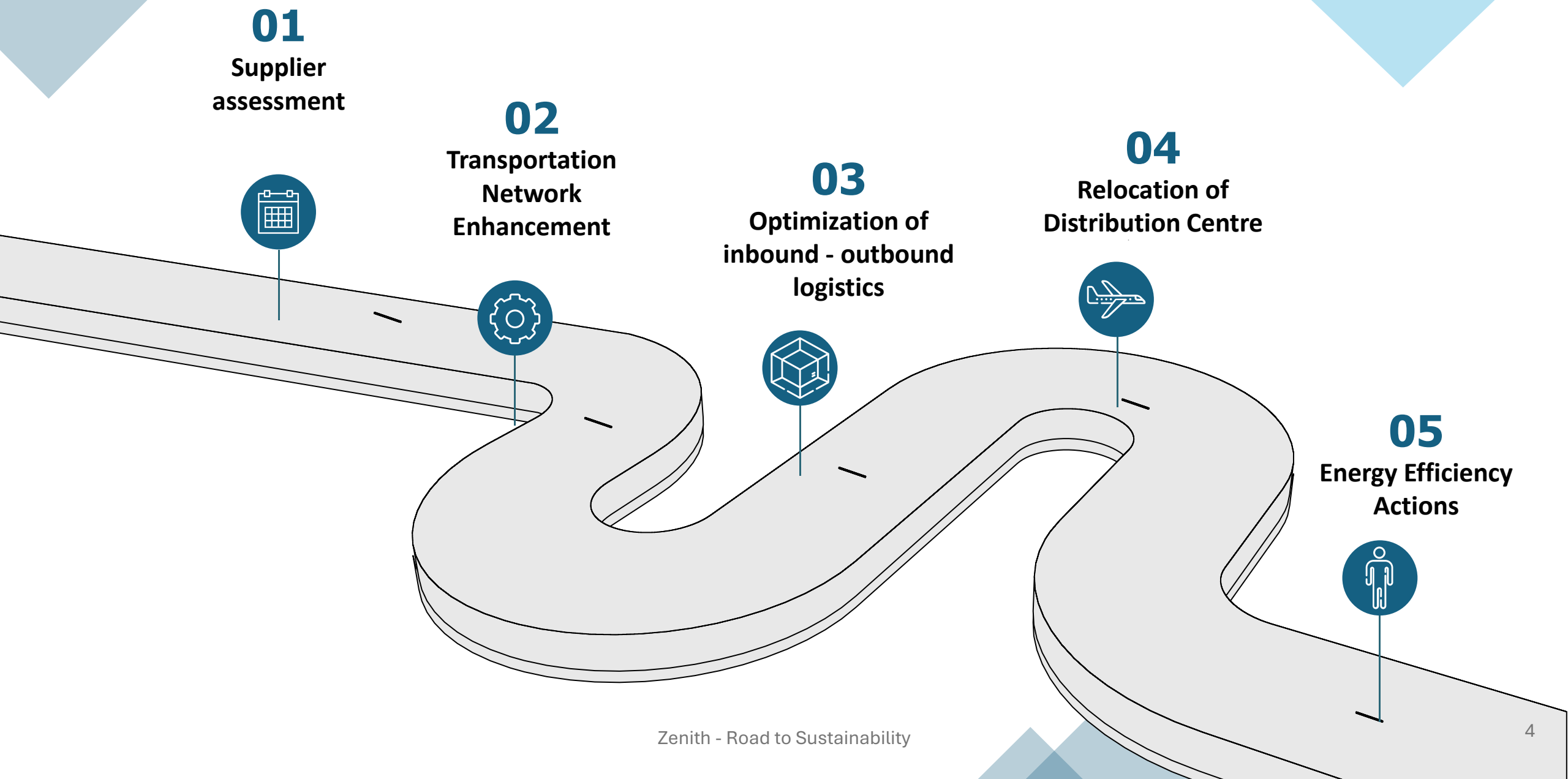
Welcome to Zenith Consulting, your trusted partner for end-to-end supply chain transformation solutions. With a deep understanding of the complexities and challenges faced by businesses today, we specialize in guiding companies through every stage of their supply chain journey.

Whether you are navigating rapid growth, overcoming operational hurdles or seeking to enhance your global competitiveness, Zenith Consulting is here to empower your supply chain success.

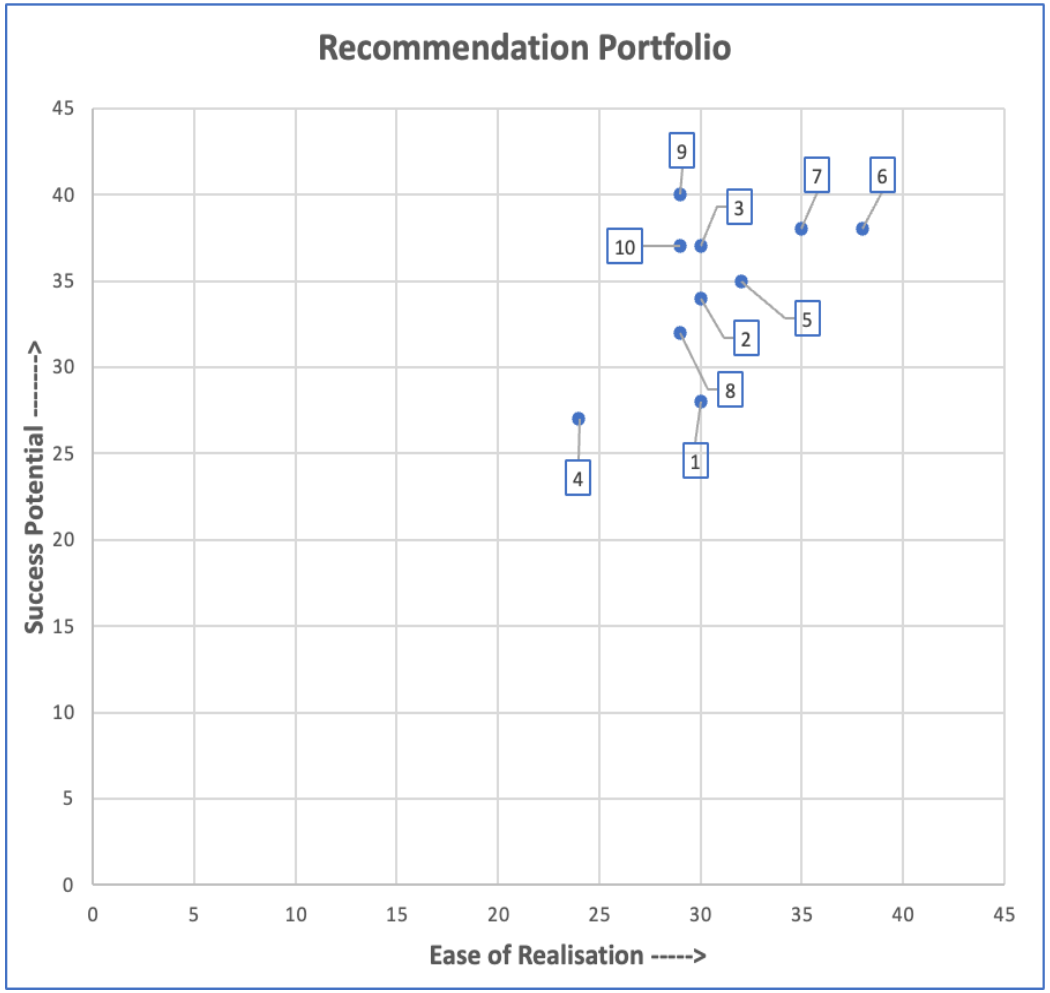
ZENITH

ROAD TO SUSTAINABILITY

Strategic Roadmap



Selection Criterion for Top 5 measures



Plotting Success Potential v/s Ease of Realization we can arrive at the most feasible recommendations			Values (0, …, 45)	
Recommendation		**Comments**	SP	ER
1	Current Supplier Assessment	Top Recommendation	**28**	**30**
2	New Supplier Development		**34**	**30**
3	Performance Metrics & Reporting		**37**	**30**
4	Circular Economy		**27**	**24**
5	Transportation Network Enhancement	Top Recommendation	**35**	**32**
6	Optimization of inbound logistics	Top Recommandation	**38**	**38**
7	Optimization of Outbound logistics		**38**	**35**
8	Warehouse Optimization		**32**	**29**
9	Relocation of Distribution Centre	Top Recommandation	**40**	**29**
10	Energy Optimization for Sustainability	Top Recommendation	**37**	**29**

Supplier Assessment

These metrics provide an overview of the supplier assessment program.

Metric	Supplier Assessment
Supplier Performance Rating	4.2 out of 5
On-time Delivery Rate (%)	95%
Product Quality Score	90%
Supplier Diversity	Increased diversity targets
Environmental Compliance	Enhanced environmental criteria
Social Responsibility	Formalized Corporate Social Responsibility program
Risk Management Procedures	Proactive risk mitigation
Supplier Relationship Management	Structured engagement plan
Technology Integration	Automated assessment platform

KPI based on Sustainability Target

1	Reducing Supply Chain Waste	(Initial Waste - Current Waste) / Initial Waste * 100%
2	Evaluate Recycling Rates	(Amount of Recycled Material / Total Material Used) * 100%
3	Measuring Greenhouse Gas Emissions/ Carbon Footprint	Total Greenhouse Gas Emissions / Total Units Produced
4	Tracking Energy Consumption	Total Energy Consumed / Total Units Produced
5	Streamline Transportation Processes	Total Transportation Costs / Total Distance Travelled
6	Impact on Ecological Systems	Use a weighted scoring system based on ecological impact assessments.
7	Track Products at Each Level	(Number of Traced Products / Total Products) * 100%
8	Water Consumption	Total Water Consumed / Total Units Produced
9	Material Quality Check for Future Value Creation	(Number of Quality Materials / Total Materials Tested) * 100%

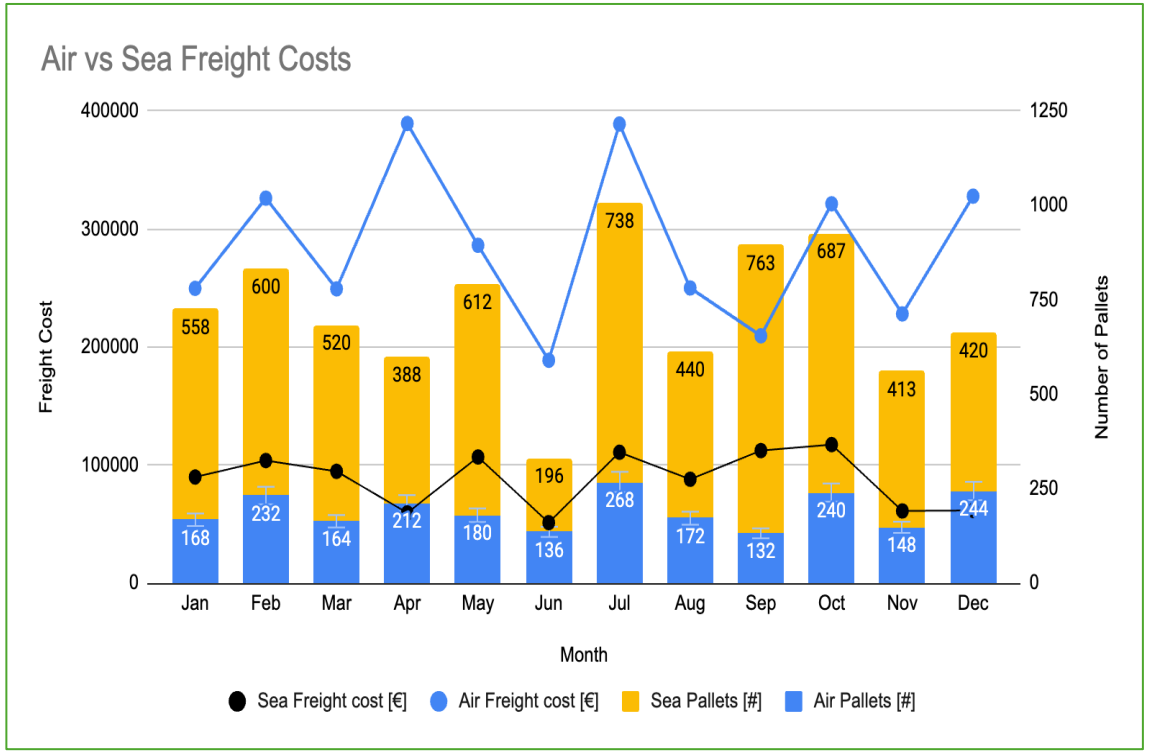
Transportation Network Enhancement

Current Challenges

- Inefficient routes & carrier selection.
- Delays at ports/customs.
- Higher inventory levels
- Higher Air freight

Objectives and Goals

- Reduce transportation costs by 20% through improved route planning and carrier selection.
- Improve delivery times by 10% by optimizing shipping processes and collaborating with carriers.
- Enhance inventory management by 15% through better forecasting and demand planning.



Resource Requirements

- Transportation Management System (TMS) software for route optimization and carrier management.
  Costs:- €10,000 - €25,000 per year (Cloud-based) or €50,000 - €100,000 one-time cost (On-premise)
- Hiring additional logistics specialists(e.g., transportation analysts).
  Costs:- € 100,000 on Manpower
  Overall Investment:- € 200,000

Transportation Network Enhancement

Potential Savings

- Reduced transportation costs due to optimized routes, carrier selection, and negotiation.
- If 50% of the air freight shipments in six months can be moved to sea freight, we can realise a profit of € 450000.

Return on Investment

Return on Investment (ROI)
•ROI = (Net Benefits / Total Costs) x 100%
•Net Benefits = € 450000
•Total Costs = € 200000
•ROI = 225% in six months.

Key Performance Indicators for Transportation

KPI	Baseline Value	Target Value	Improvement
Transportation Cost	$100	$80.00	20.00%
Delivery Time	5 days	4.5 days	10.00%
Inventory Holding Cost	$20/unit/month	$17.00/unit/month	15.00%
Inventory Level	1000 units	850 units	15.00%

Optimization of Inbound and Outbound logistics

Current Challenges

Transport Cost Optimization:-

- Higher use of air freight as compared to sea freight.
- Higher travel distances to the end customer.
- High cost of maintenance for own fleet of delivery vehicles.

Environmental Impact:-

- Air freight is associated with a significantly higher carbon footprint compared to sea freight due to higher fuel consumption per unit of cargo transported.
- Higher CO2 emissions due to the higher number of miles travelled from the existing DC to the customer.

Objectives and Goals

- Use third party delivery network for delivering goods to customer.
- Use of railway network for the inbound deliveries.
- By effect of relocation, the distance travelled to customers will reduce drastically owing to cost savings.
- Get rid of the 160 vans owned by the company giving us an approx income of 2.4 million.

Resource Requirements

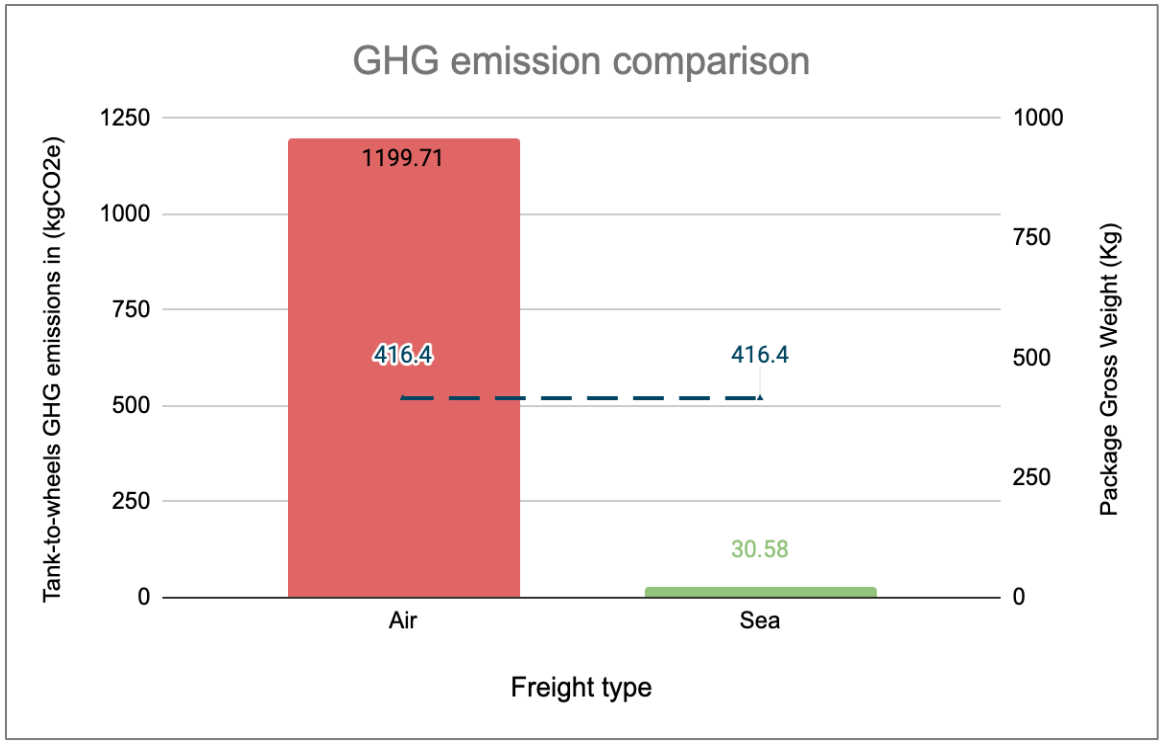
- Own fleet of Electric vans for urgent and priority customer deliveries. Approximately 10 vehicles.
  Costs:- € 700,000
  Overall Investment:- € 700,000

Optimization of Inbound and Outbound logistics

- Reduced GHG emissions as sea freight produces upto 95% less KgCO2e compared to air freight.
- Considerable savings can be realised if outbound logistics are made using third party delivery network.

Potential Savings

Return on Investment



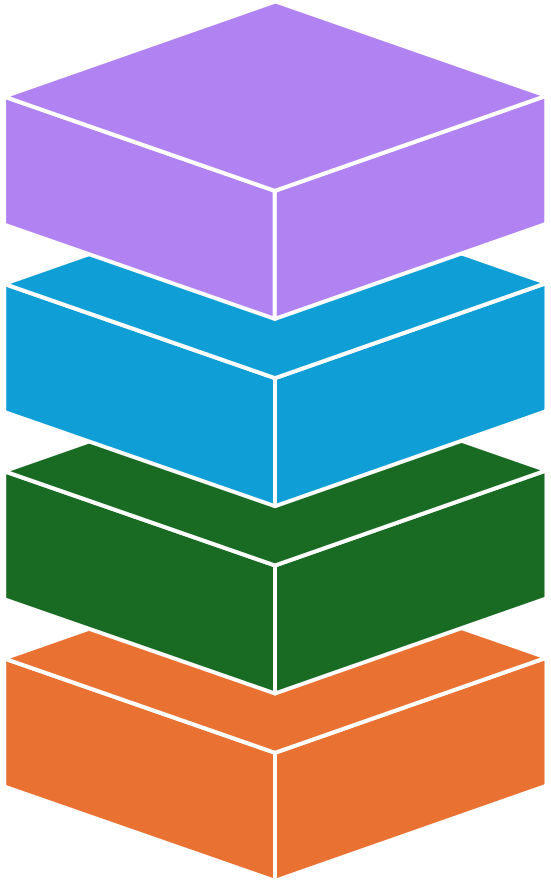
Return on Investment (ROI)

•ROI = (Net Benefits / Total Costs) x 100%
•Net Benefits = € 2400000
•Total Costs = € 700,000
•ROI = 342%.

Energy optimization for Sustainability

LEDs offer significant energy cost reductions compared to traditional lighting.
We can expect a **50% to 80% reduction** in energy consumption for warehouse lighting.

	Resource Requirements	Electricians, LED lighting fixtures, potential rewiring depending on existing infrastructure.
€55,000	Investment	(assuming a 10,000 sq. m warehouse)
€11,000	Potential Savings	Annual energy cost savings of €11,000 (based on an estimated 50% reduction in energy consumption)
5 Years	Return on Investment (ROI)	(considering potential installation costs and ongoing maintenance)



Long-Term Opportunities, Risks and Benefits

Opportunities

- Continuous improvement: Leverage data from KPIs (Q4) to identify further optimization opportunities.
- Attract new customers: Sustainability focus can attract environmentally conscious businesses and consumers.
- Regulatory compliance: Proactive sustainability measures can prepare ELEX for upcoming environmental regulations.

Risks

- Unexpected implementation costs: Thorough planning and vendor selection are crucial to minimize cost overruns.
- Employee resistance to change: Training and communication are essential to ensure employee buy-in for new processes.
- Potential disruption during implementation: Careful planning and phased implementation can mitigate operational disruptions.

Sustainability Benefits

- Reduced Energy Consumption: Both proj ects contribute to lower energy use, leading to a decreased carbon footprint.

Business Benefits

- Cost Savings: Project offers significant cost savings through reduced energy expenditure
- Enhanced Brand Image: Demonstrating sustainability efforts can attract environmentally conscious customers.

Monitoring for Energy Consumption

Metric 1

01

Energy Consumption per Square Meter per
year Kwh/ SqM < 20 Kwh/sqm

02

Formula: (Previous Year's kWh Consumption per sqm - Current Year's kWh Consumption per sqm)

03

Previous Year's kWh consumption per sqm = 320,000 kWh / 10,000 sqm = 32 kWh/sqm

04

After implementing LED lighting, considering the current year's total kWh consumption is 160,000 kWh.

05

Expecting current Year's kWh consumption per sqm 16 kWh/sqm

06

Adding some tolerance value and rounding to 20 KWh/ sqm

Relocation of Distribution Center

Most of the outbound shipments travel a distance of 400+kms which is not an ideal case considering the size of Germany. Relocate the existing DC to Fulda which is centrally located in Germany. It is well-connected to air and rail transportation, serving as a logistics hub for the country.

- Site Selection : Fulda, Zip code: 36037
- Resource Requirements: Land, Civil Construction Contractor, Electricians, Commodities
- Site Area :- 10000 Sqm
- Personnel availability : -15 staff x 6 days (8 hrs shift)
- Investment: €1,00,85,500 (Listing major costs)
  Land Rent cost: €10500 (Assuming 3% Rental Yield){Land Cost €350,000 (Assuming a 10,000 sq. m warehouse)}
  Construction Costs: €10,000,000 (Building costs in Germany average around €1,000 per sqm)
  Warehouse Management cost: €15000
  Material Handling Equipment: €50000
  Additional Consideration: €10000
- Potential Savings: €300,000 (Saving in Transportation)
- Return on Investment (ROI): 3 Years

Relocation of Distribution Center

Technology	Warehouse Managment System (WMS)	€10k-25k
	Material Handling Equipment: (a) Forklifts (b) Automated Guided Vehicles (AGVs) (c) Pick-to-Light Systems (d) Automated Storage and Retrieval Systems (AS/RS)	€100K
	Security System like camera, acess control system,alarm detector	€5K
Material	Storage Solutions: (a) Pallet Racking (b) Shelving (c) Bins and Containers	€90K
Information Flow:	Real-time Data Visibility, Integration with Other Systems, Barcode Scanning, Electronic Data Interchange (EDI)	€10-20K

Recommendation Selection Criterion

	Recommendations	Description	Impact on Sustainability
01	Current Supplier Assessment	Evaluating the supplier based on environmental impact, compliance, employment practices, reporting, supplier labor initiatives, and green innovation.	• Responsible sourcing Leads to good labor practices Increases the use of recycled packaging
02	New Supplier Development	Identify supplier competitors for diversification and competitive pricing, prioritize those with lower environmental impact, financial stability and scalability.	• Leads to a partnership with sustainable suppliers who are aligned with your sustainability goals
03	Performance Metrics & Reporting	Establish KPIs with specific metrics for environmental, social, economic impact, and implement tracking and reporting for transparency and stakeholder engagement.	• Drives continuous sustainability improvement. Improves stakeholder engagement.
04	Circular Economy	Implement buyback programs for customers' old electronic devices, ensuring responsible recycling, material recovery, and minimal environmental impact.	• Promotes sustainable waste management
05	Transportation Network Enhancement	Collaborate with freight partners prioritizing in initiatives like carbon offset programs. Consolidate shipments for efficiency and emissions reduction.	• Reduced Co2 emissions Lower overall transportation costs.
06	Optimization of inbound logistics	Shift inbound shipments from air to sea freight for less time-sensitive deliveries, leveraging rail and green trucks for sustainable distribution.	• Fewer emissions per ton-mile than road transport. Sea freight has a lower carbon footprint than air.
07	Optimization of Outbound logistics	Partner with sustainable logistics providers and maintain a fleet of green vehicles for critical deliveries.	• Reduced Co2 emissions associated with transportation.
08	Energy Efficiency Actions in Existing DC	Analyze DC consumption; replace lighting with LEDs, use natural ventilation, explore solar panels feasibility, and consider sustainable materials for pallets/racking.	• Reduces energy consumption Promotes responsible resource management
09	Relocation of Distribution Centre	Relocate the existing DC to Fulda which is centrally located in Germany. It is well-connected to air and rail transportation, serving as a logistics hub for the country.	• Reduces the overall miles travelled to deliver to the customer thereby reducing the CO2 emissions.
10	Warehouse Optimization for Sustainability	Organize space to prevent congestion, label items clearly for efficient retrieval. Implement scanners, RFID tags, IoT systems, and AI for real-time data collection, analysis, and inventory management.	• Enhances the overall efficiency with automation, lowering waste and environmental impact.

Recommendation Portfolio

SUCCESS POTENTIAL: To what extent does the idea effectively solve our problem?
EASE OF REALIZATION: To what extent is the idea easy to realize for us?

1. Criterion specified to evaluate the Top 10 recommendations with regard to their SUCCESS POTENTIAL and the EASE OF REALIZATION.
2. Assign a weight factor Wi for each criterion

	Evaluation criterion (C)		*Description*	*Weight $w_i = w(c_i)$* (1: low, 2: medium, 3: high)
Success potential (SP)	C_{SP1}	Impact on Key Performance Indicators (KPIs):	Evaluate the potential impact of each measure on critical KPIs such as **cost reduction**, **lead time improvement**, and **sustainability metrics**.	3
	C_{SP2}	Scalability and Adaptability:	Consider the scalability and adaptability of each measure to accommodate **future growth**, **changes in market conditions**, and **emerging trends**.	3
	C_{SP3}	Alignment with Strategic Objectives:	Assess the degree to which each measure aligns with the company's strategic objectives and long-term vision for the supply chain.	2
Ease of realization (ER)	C_{ER1}	Resource Requirements:	Evaluate the resources such as **financial**, **human**, **technological**, etc. required to implement each measure effectively.	3
	C_{ER2}	Complexity and Integration:	Assess the complexity of implementing each measure and its compatibility with **existing systems**, **processes**, and **infrastructure**.	3
	C_{ER3}	Stakeholder Support and Collaboration:	Consider the level of stakeholder support and collaboration needed to implement each measure successfully.	2

Recommendation Portfolio

SUCCESS POTENTIAL: To what extent does the idea effectively solve our problem?
EASE OF REALIZATION: To what extent is the idea easy to realize for us?

How do the recommendations fit to our assessment criteria?

1. Set **fitting coefficient** $s_{j,i}$: The fit of $idea_j$ to $criterion_i$ is: 0 = extremely poor, …, 5=extremely high
2. Multiply $s_{j,i}$ with the corresponding weight factor w_i
3. Sum up row values $s_{j,SPi}$*w_{SPi} to give the score for the success potential SP_j of $idea_j$ (ER_j analog)

		Success potential							Ease of realization						
Top 10 Recommendations		$s_{j,SP1}$	sj,SP1*wSP1	sj,SP2	sj,SP2*wSP2	sj,SP3	sj,SP3*wSP3	Score SP SPj = $sj,i*wi$	sj,ER1	sj,ER1*wER1	sj,ER2	sj,ER2*wER2	sj,ER3	sj,ER3*wER13	Score ER ERj = $sj,i*wi$
01	Current Supplier Assessment	3	9	3	9	5	10	**28**	4	12	4	12	3	6	**30**
02	New Supplier Development	4	12	4	12	5	10	**34**	4	12	4	12	3	6	**30**
03	Performance Metrics & Reporting	5	15	4	12	5	10	**37**	4	12	4	12	4	8	**30**
04	Circular Economy	3	9	4	12	3	6	**27**	3	9	3	9	3	6	**24**
05	Transportation Network Enhancement	5	15	4	12	4	8	**35**	4	12	4	12	4	8	**32**
06	Optimization of inbound logistics	5	15	5	15	4	8	**38**	5	15	5	15	4	8	**38**
07	Optimization of Outbound logistics	5	15	5	15	4	8	**38**	4	12	5	15	4	8	**35**
08	Warehouse Optimization for Sustainability	3	12	3	12	4	8	**32**	3	9	4	12	4	8	**29**
09	Relocation of Distribution Centre	5	15	5	15	5	10	**40**	3	9	4	12	4	8	**29**
10	Energy Optimization for Sustainability	5	15	4	12	5	10	**37**	3	9	4	12	4	8	**29**

Resources

Resources for Energy optimization

- https://fireflier.com/the-best-number-of-warehouse-lights/
- https://formulare.get-energy.de/?dstr=check24net&product=strom&kwh=654000&branche=8&leistung=&plz=36100&ort=Petersberg
- https://www.check24.net/strom-vergleich/?c24pp_zipcode=36100&c24pp_city=Petersberg&c24pp_totalconsumption=2500&c24_calculate=
- https://ctprodstorageaccountp.blob.core.windows.net/prod-drupal-files/documents/resource/public/Warehousing-and-logistics-guide.pdf
- https://www.meteorspace.com/25-warehouse-energy-consumption-statistics-you-need-to-know/
- https://ctprodstorageaccountp.blob.core.windows.net/prod-drupal-files/documents/resource/public/Warehousing-and-logistics-guide.pdf
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- https://www2.deloitte.com/content/dam/Deloitte/de/Documents/energy-resources/Deloitte_Sustainable%20Value%20Chain%20Transformation.pdf
- Location example - https://www.frasersproperty industrial.com/en-au/property/all.html?gad_source=1&gclid=Cj0KCQjwudexBhDKARIsAI-GWYXzGZNZkZxIcp-0yaLJdU9_kAJQAxYdfRG-1GGSv4G6OFpvMEQfpH8aArqkEALw_wcB&gclsrc=aw.ds
- https://www.statista.com/statistics/527832/warehouse-primary-rent-cost-logistics-market-germany-europe/
- https://eu.statesmanjournal.com/story/money/business/development/2024/01/03/amazon-completing-internal-structure-robots-at-woodburn-warehouse/72034943007/
- U-shape:
  https://www.webstaurantstore.com/article/683/warehouse-setup-guide.html
- About warehouse and technology details for selection.
  What is Warehouse Optimization & 12 Ways To Optimize Warehouse (hopstack.io)